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# Assessment of muscle volume using magnetic resonance imaging (MRI) in football players after hamstring injuries

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**ABSTRACT**

Muscle injuries of the hamstrings are among the most frequent in football and a main cause for significant time away from training and competition. The purpose of this study was to prospectively evaluate the loss of muscle volume in recreational football players three and six weeks after initial trauma. We hypothesized that significant muscle volume loss occurs within 6 weeks after the initial injury event. Twenty recreational football players (mean-age=25 ± 4years; mean-height=181 ± 8cm; mean-weight=81 ± 10kg) with type3a (minor partial muscle tear) and type3b (moderate partial muscle tear) injuries were included. Muscle volume was determined using established methods for the hamstrings and the quadriceps femoris muscle within 3 days and after 3 and 6 weeks following the initial injury. The injured hamstrings lost 6.5% (mean=64 $cm^3$(95%CI=31–98 $cm^3$), $p<0.001$), the healthy hamstrings lost 2.1% (mean=21 $cm^3$(3–44 $cm^3$), $p=0.096$) of muscle volume after six weeks. The quadriceps in the injured leg lost 3.8% (mean=78 $cm^3$(51–104 $cm^3$), $p<0.001$) and 4.5% (83 $cm^3$ (45–121 $cm^3$), $p<0.001$) in the healthy leg. Muscle volume loss inversely correlated with activity levels in the healthy leg for the quadriceps (r=0.96 (0.90–0.98); $R^2=0.92$; $p<0.001$) and the hamstrings (r=0.72 (0.40–0.88); $R^2=0.51$; $p<0.001$), as well as the quadriceps in the injured leg (r=0.70 (0.37–0.87); $R^2=0.49$; $p<0.001$), but not the injured hamstrings. Muscle volume ratio of hamstrings to quadriceps in the control limb was 0.52 ± 0.06 and 0.53 ± 0.06 in the injured leg. The rehabilitation period of six weeks did not have a relevant negative or a positive effect on ratios. Significant muscle volume loss in the upper thigh occurs in recreational soccer players within three, and within six weeks after a hamstring injury and lies between 2% and 7%.

**KEYWORDS**
Muscle volume measurements; hamstring injuries; muscle volume loss; rehabilitation

## Highlights

- This study found that significant muscle volume loss occurs within six weeks after sustaining hamstring muscle group injuries.
- Additionally, most of the non-professional football players in the present study demonstrated low hamstring-to-quadriceps volume ratios in both the injured, as well as the heathy leg. Low hamstring-to-quadriceps volume ratios may provide an opportunity to focus on muscle hypertrophy strategies to help reduce the risk for muscle imbalances and thus future injury.
- After three weeks of resting, reduced training activity, compared to the training activity before the injury, seems to be sufficient to preserve 99% of the remaining muscle volume in the quadriceps and as well as the healthy hamstrings. The number of training-minutes correlates statistically significant inversely with the muscle volume lost in a rehabilitation period.

## 1. Introduction

Hamstring muscle injuries frequently occur among recreational (Liu, Garrett, Moorman, & Yu, 2012; Seymore, Domire, DeVita, Rider, & Kulas, 2017) and elite athletes (Ekstrand, Hägglund, & Waldén, 2011; Ekstrand, Waldén, & Hägglund, 2016), with a high prevalence in sports characterized by sudden acceleration and maximal sprinting, such as football (Arnason et al., 2004; Arnason, Gudmundsson, Dahl, & Jóhannsson, 1996; Hawkins & Fuller, 1999; Woods et al., 2004). The mechanism of injury typically involves forced flexion of the hip with the ipsilateral knee in extension like in

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high speed running (Cohen, Rangavajjula, Vyas, & Bradley, 2012; Lempainen et al., 2015). Hamstring muscle injuries cause significant time away from training and competition and may therefore have significant negative effects on sports careers and quality of life (Bennell et al., 1998; Verrall, Slavotinek, Barnes, Fon, & Spriggins, 2001; Woods et al., 2004).

The harmonious contraction and relaxation of selected muscle fibers generates movement in the human body. Recent data demonstrated that the strength of a muscle is linearly related to its physiological cross-sectional area, which is obtained from the volume of the muscle divided by its fiber length (Fukunaga et al., 2001; Suchomel, Nimphius, Bellon, & Stone, 2018). Hence, a strong relation between muscle volume and strength may be expected (Baxter & Piazza, 2014; Evangelidis, Massey, Pain, & Folland, 2016; Fukunaga et al., 2001; Gadeberg, Andersen, & Jakobsen, 1999; Kellis, Sahinis, Dafkou, Ellinoudis, & Galanis, 2021). It was recently recognized that muscle strength imbalances between the quadriceps femoris muscle and the hamstring muscle group (ratio < 0.61) may significantly increase the likelihood of sustaining a hamstring injury up to 5 times (Croisier, Ganteaume, Binet, Genty, & Ferret, 2008; Green, Bourne, van Dyk, & Pizzari, 2020; Orchard, Marsden, Lord, & Garlick, 1997). However, the hamstring to quadriceps ratio has so far only been described in strength values and not in muscle volume.

Muscle loss after an injury-related period of abstinence from training is perceived as a main cause for strength deficits and thus, increased time away from competition (Kilroe, Fulford, Jackman, van Loon, & Wall, 2019). While recent findings showed that muscle strength levels may be maintained during 3 weeks of absence from training, significant muscle volume loss of almost 1% per day during complete immobilization may occur (Dirks et al., 2016; Kilroe et al., 2019; McMaster, Gill, Cronin, & McGuigan, 2013). However, there is limited data about the typical loss of muscle volume after muscle injuries over a longer period. It is still speculative whether atrophy or muscle size imbalance increases the risk for injury. However, it is of clinical relevance as rehabilitation protocols should aim for realistic expectations of muscle recovery (Ramos et al., 2017).

The purpose of this study was to gain a better understanding of the typical muscle volume change in the upper thigh after a hamstring injury. This study was performed to (1) evaluate the loss of muscle volume in recreational football players over the course of six weeks after sustaining muscle injuries to the hamstring muscle group compared to the healthy, non-injured leg, (2) to investigate the muscle volume ratio between the quadriceps femoris muscle and hamstrings muscle group for both limbs over the course of six weeks and (3) to evaluate the correlation of activity after a period of non-training with change in muscle volume.

The authors expect significant differences in muscle volume changes after three and after six weeks when comparing the injured to the control limb, i.e. more volume loss in the affected muscles.

## 2. Materials and methods

### 2.1 Study design

Approval from the institutional review board of the medical faculty and local ethics committee (**41/18**) was obtained prior to performing this study. Informed consent was obtained from all individual participants included in the study. Recreational football players sustaining hamstring muscle injuries between 01/2018 and 03/2019 were prospectively enrolled. Subjects eligible for study inclusion had to be 18 years or older with a hamstring injury sustained during a football-match or practice. The injury was confirmed as a fresh injury in the hamstring muscle group by magnetic resonance imaging (MRI) by a specialised radiologist within 3 days of injury and classified according to the Munich classification (Mueller-Wohlfahrt et al., 2013). Participants had to be available for a radiographic follow-up 3 and 6 weeks after the initial trauma. Football players with multiple injuries in muscles other than the hamstring muscle group, complete proximal or distal hamstring (avulsion) injuries, previous hamstring injuries within the last 12 months, as well as any other concomitant injuries to the lower extremity were excluded from this study. Only male patients were included to achieve a homogenous data set.

### 2.2 Rehabilitation protocol

For the complete rehabilitation period, the players were mobile, without the use of walking aids like crutches. The rehabilitation protocol consisted of two phases. In the first phase, all participants were advised to rest for at least three weeks and not engage in any strenuous activity, upon being diagnosed with a hamstring injury. They followed their normal daily life routines through the complete period of rehabilitation. Three weeks after sustaining the injury recommendations were adjusted individually, to reflect the amount of structural damage still present in the MRI-findings, as well as the individual pain level. Players documented the number of minutes invested in rehabilitation activity for each week from week four to week six. Activities included running, cycling, fitness and football practice. No

restriction of minutes was given to reflect the current clinical practice, as physiotherapists treat non-professional athletes based on their individual pathology, presentation, and goals individually.

### 2.3 MRI-Analysis

MRI imaging was performed on a 3-Tesla whole body Magnet Resonance (MR) scanner (Prisma Fit, Siemens, Erlangen, Germany). The protocol included T1-tirm (turbo inversion recovery magnitude) sequences taken coronar bilateral and transversal bilateral, PD (Proton density), TSE (ultra turbo spin echo) and FS (fat-sat technique) sequences taken coronar and parasagittal with a layer thickness of two millimeters angulated over the detected lesion. No contrast was used. MRI were performed with the players lying prone. A body coil was laid over the region of interest of the affected and non-affected limb. Most players return to sport three weeks after a hamstring injury (Connell et al., 2004; Reurink et al., 2014; Warren, Gabbe, Schneider-Kolsky, & Bennell, 2010). After six weeks some structural damage can still be expected in the MRI (Connell et al., 2004). Therefore, MRIs were performed at time 0 ($MRI_0$), at 3-weeks ($MRI_3$) and 6-weeks ($MRI_6$) after sustaining the injury. MR images were analyzed by one orthopedic trauma surgeon under supervision of a musculoskeletal radiologist consultant, using a DICOM medical imaging viewer (Horos v3.0, Horos project, Annapolis).

### 2.4 Muscle volume measurements

Muscle volumes were determined for both limbs using manual measurement in the bilateral transversal t2-sequence. Volumes of the hamstring muscle group and the quadriceps were quantified individually and combined for each leg, respectively. The segments were measured digitally using the exact same transversal layers with the same number of slices, for each of the three MRIs for consistency in each player. For MRIs that did not visualize the complete muscle, the complete data set was excluded from this study. In each slice, the outlines of the hamstrings and quadriceps of each leg were outlined from the superior border of the patella to the point where no muscle of the group was distinguishable any more to calculate the area of the outlined muscle (figure 1). In the first three weeks resorption of edema in the injured hamstrings occurred, which could not be factored out in the measurement. To assess the intra- and inter-investigator reproducibility in our study, all of the slices from the first MRI in the healthy legs were outlined a second time on different days by the same investigator, and on another occasion by a second investigator.

Subsequently, each subject's three-dimensional anatomy of the quadriceps and hamstring muscles was digitally reconstructed. Muscle volume in $cm^3$ was calculated for the hamstrings muscle group and the quadriceps for each leg, respectively by summation of the volumes of each slice.

### 2.5 Statistical analysis

Intra- and inter-investigator reproducibility in this study was determined using the intraclass correlation coefficient (ICC) and the coefficient of variation (CV) for the quadriceps and the hamstring muscle group muscle volume in the healthy leg. We compared the muscle volume from the second and third MRI with the first measurement to determine not only the absolute muscle volume loss in $cm^3$, but the relative muscle loss in percent for each player as well. Percent change ($\Delta\%$) against the initial muscle volume was determined in each muscle group for each individual leg by calculating the relative muscle volume compared to the volume measured in the first MRI. For each muscle group, the mean ($M$) and standard deviation ($SD$) of $cm^3$ as well as the percent difference versus the initial volume of the respective leg were calculated across all individuals (n=20). The Shapiro–Wilk test was used to determine if the data followed normal distribution. All the data followed normal distribution.

A one-way ANOVA using the Geissner-Greenhouse correction was used for each muscle group on each leg comparing the absolute as well as the relative volume after three weeks and six weeks with the volume measured in the first MRI ($MRI_0$). Bonferroni corrections were used as statistical hypothesis testing for multiple comparisons. Tests were calculated two-tailed using a 95% confidence level. $P$-values $< 0.05$ were considered significant. Correlation coefficients between muscle volume change in percent and the minutes of activity documented by the players were calculated using Pearson's r with two tailed $p$-values. The following descriptors were used to assess the strength of the correlations: "low" = below 0.39, "moderate" = 0.40–0.59, "strong" = 0.60–.79, and "very strong" = 0.90–1.00.

To compare both legs, the differences in muscle volume change between the healthy leg and injured leg after three and six weeks were computed for the quadriceps and for the hamstrings. A paired t-test was then used to determine significant differences in muscle volume changes between both legs. Hamstring to quadriceps volume ratios were calculated to determine the level of imbalance between these two

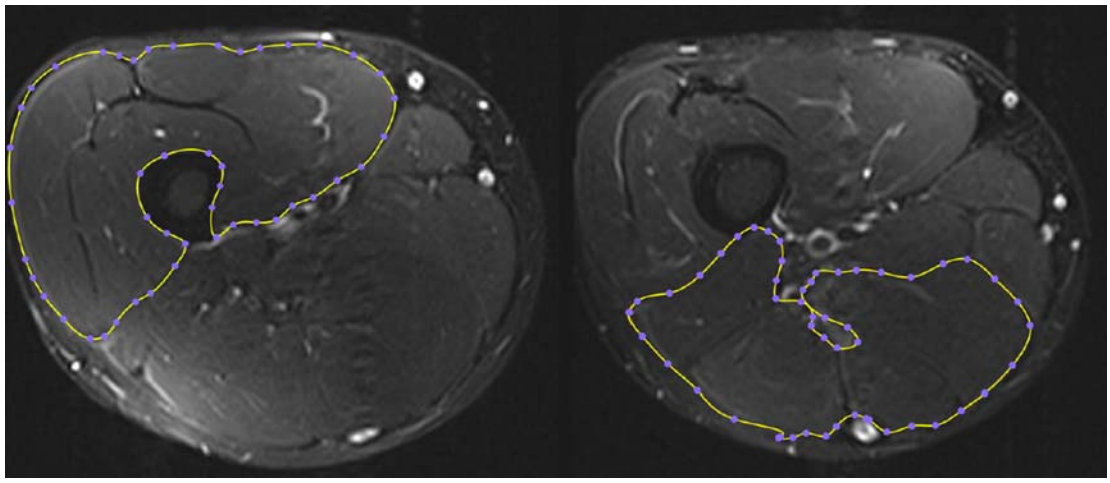

**Figure 1.** The Outlines of the cross-sectional area of the quadriceps muscle (left) and the hamstring group (right) are presented in a transversal layer of a Magnet Resonance Image (MRI). Hereby the volume for each muscle in the slice was calculated.

muscle groups in the injured and uninvolved limb for $MRI_6$. All statistical analysis was performed using Prism Graphpad 8 (GraphPad Software, CA, San Diego).

## 3. Results

### 3.1 Subjects

Of the twenty-six players, six were excluded from the study, as the displayed sector of the upper thigh did not image the complete muscle. Final data analysis included twenty, male recreational football players (age: 25 years ± 4, height: 181cm ± 8, weight: 81kg ± 10). All athletes carried out football specific training including competition and matches for a total of three to six hours a week (mean = 315 ± 75 min) on a regular basis before sustaining the injury. Twelve players had a type 3a (minor partial muscle tear) while eight players had a type 3b (moderate partial muscle tear) hamstring muscle injury according to the Munich classification (Mueller-Wohlfahrt et al., 2013). The injury occurred in the dominant leg in eighteen cases. The MRIs of the upper thigh were conducted within the first three days after sustaining the injury ($MRI_0$; 45 ± 15 h), three weeks post trauma ($MRI_3$; 21 ± 2 days) and six weeks post trauma ($MRI_6$; 43 ± 4 days). Each player individually documented rehabilitation activities (cycling, swimming, running, fitness, football) from week 4 to week 6 with a total of 91 ± 77 min in week 4, 134 ± 100 min in week 5, and 107 ± 73 min in week 6.

### 3.2 MRI-Analysis

#### 3.2.1 Intra- and inter-investigator reproducibility

The mean muscle volumes for the three measurements for the healthy quadriceps were 1890 ± 162 cm3 (95% CI = 1814–1965 $cm^3$) and 1891 ± 186 cm3 (1803–1980

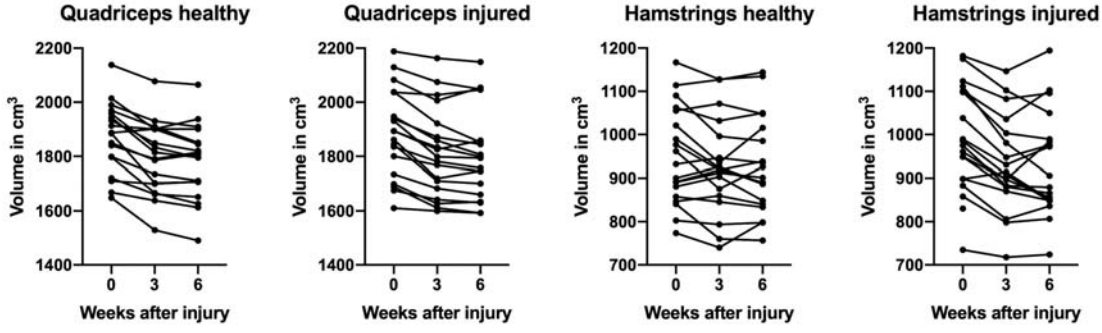


**Figure 2.** Muscle volumes (*y*-axis) for each player over the course of three and six weeks (*x*-axis) for the quadriceps muscle in the healthy leg (Quadriceps healthy), the quadriceps muscle in the injured leg (quadriceps injured), the hamstring muscle group in the healthy leg (hamstrings healthy) as well as the hamstring muscle group in the injured leg (hamstrings healthy) individually. The trend shows a larger loss of muscle volume in the first three weeks compared to from week 4 to week six. Overall almost all players lost muscle volume compared to the starting point.

$cm^3$) by the first investigator and 1890 ± 162 cm3 (1815–1966 $cm^3$) by the second investigator. The mean muscle volumes for the three measurements for the healthy hamstring group were 964 ± 124 cm3 (905–1021 $cm^3$) and 962 ± 125 cm3 (903–1021 $cm^3$) by the first investigator and 960 ± 121 cm3 (904–1017 $cm^3$) by the second investigator.

Excellent intra- and inter-investigator reproducibility was observed for the quadriceps (ICC = 0.974 (mean difference = 1.75 ± 48.56 cm3 (−20.98–24.48); $p = 0.874$) and 0.989 (mean difference = 0.85 ± 23.94 cm3 (−10.35–12.05); $p = 0.876$) respectively; CV = 0.7% ± 0.8 and 0.6% ± 0.3 respectively) as well as the hamstrings (ICC = 0.990 (mean difference = 1.00 ± 18.09 cm3 (−9.47–7.47); $p = 0.807$) and 0.988 (mean difference = 2.80 ± 19.45 cm3 (−11.90–6.30); $p = 0.527$) respectively, CV = 0.8% ± 0.5 and 0.9% ± 0.5 respectively). This is in line with previous studies, that demonstrated an coefficient of variation of under 1% for muscle volume measurements obtained from MR image data (Nordez et al., 2009; Tracy et al., 2003).

#### 3.2.2 Muscle volume loss

The muscle volumes for each player and muscle group over the course of the 6 weeks are presented in figure 2.

Compared to time 0 ($MRI_0$), the injured hamstring muscle group showed a mean decrease in muscle volume of 6.7% after three weeks. This includes the resorption of edema in the injured hamstrings, which was observed during the first 3 weeks. The muscle volume loss of the healthy control hamstrings was 2.2% in the mean compared to time 0. The quadriceps in the injured leg presented a mean muscle volume loss of 3.5% and in the healthy leg with 3.9% over the same period. Mean differences in volume changes between both legs were 46 $cm^3$ ± 45 (95%CI = 25 −67; $p < 0.001$) for the hamstrings and 6 $cm^3$ ± 47 (−26 – 12; $p = 0.492$) for the quadriceps muscles (table 1).

Compared to time 0 ($MRI_0$), the injured hamstring muscle group showed a mean muscle volume decrease of 6.5% after six weeks. Mean muscle volume loss of the healthy control hamstring group was 2.1%. The mean muscle volume loss of the quadriceps occurred with 3.8% in the injured leg and 4.5% in the healthy leg. The mean difference in muscle loss between the injured and healthy leg after six weeks was 43 $cm^3$ ± 48 (21 – 66; $p < 0.001$) for the hamstrings and 5 $cm^3$ ± 41 (−27 – 16; $p = 0.603$) for the quadriceps (table 2).

The mean muscle volume ratio of the hamstrings to the quadriceps in the control limb was 0.52 ± 0.06 at the starting point. No relevant change was noted within three weeks (0.52 ± 0.06; $p = 0.984$) and six weeks (0.52 ± 0.06; $p = 1.000$).

The injured leg showed a slight decrease in hamstring to quadriceps ratio from 0.53 ± 0.06–0.51 ± 0.07 after three (0.02 ± 0.00 (0.01 – 0.02); $p = 0.019$) and 0.51 ± 0.07 (0.01 ± 0.1 (0.00 – 0.03); $p = 0.154$) after 6 weeks. No relevant difference between the control and the injured limb was noted (0.01 ± 0.00 (0.0 – 0.01); $p = 0.471$).

### 3.3 Correlation of activity and muscle volume change

The muscle volume change in percent showed a very strong correlation with the activity of the players in minutes from week four to week six in the healthy control limb for the quadriceps (r = 0.96 (95%CI = 0.90–0.98); $R^2 = 0.92$; $p < 0.001$), and a strong correlation for the hamstrings (r = 0.72 (0.40–0.88); $R^2 = 0.51$; $p < 0.001$), as well as the quadriceps in the injured leg (r = 0.70 (0.37–0.87); $R^2 = 0.49$; $p < 0.001$). There was low correlation between the injured hamstrings and muscle volume gained with activity during this time. (r = 0.39 (−0.66–0.71); $R^2 = 0.15$; $p = 0.091$).

## 4. Discussion

The most important finding of this study was that recreational football players that do not engage in strenuous activities within a period of three weeks following a hamstring-injury lose between 2 and 7% muscle volume in the hamstrings and quadriceps muscle in the mean. Reduced training activity of around 111 min ± 85 per week was then sufficient to preserve 99% of the remaining muscle volume from week three to week six. Thus, compared to the usual training level, a reduced training activity compared to the training level before injury may be enough to prevent substantial further muscle volume loss in the proximal lower limbs, mostly depending on players pain level and grade of injury (Brukner, 2015; Lempainen et al., 2015).

Prior to the study, the authors expected significant differences in muscle volume changes when comparing the injured to the control limb, i.e. more volume loss in the affected muscles. The results from this study are similar to recent findings, however, supporting a physiological asymmetry of around 10% (Kulas et al., 2018). No relevant difference in volume loss was observed between the quadriceps of the injured leg and the healthy leg after 6 weeks.

Thus, with reduced training activity, muscle volume loss in the quadriceps between the injured and healthy leg are comparable within the first 6 weeks after sustaining hamstring injuries.

**Table 1.** MRI-Analysis after 3 weeks.

| *Muscle group* | *Mean volume ($MRI_0$) in $cm^3$ + SD* | *Mean volume ($MRI_3$) in $cm^3$ + SD* | *Mean volume loss in $cm^3$* | *95% CI* | *Mean pc in $MRI_3$ + SD* | *p-value* |
|---|---|---|---|---|---|---|
| Quadriceps injured leg | 1910 ± 191 | 1843 ± 198 | 67 | 45 – 88 | 3.5 ± 2.1 | < 0.0001 |
| Quadriceps healthy leg | 1890 ± 162 | 1817 ± 166 | 73 | 47 – 99 | 3.9 ± 2.6 | < 0.0001 |
| Hamstrings injured leg | 1004 ± 144 | 938 ± 143 | 67 | 46 – 87 | 6.7 ± 4.1 | < 0.0001 |
| Hamstrings healthy leg | 964 ± 124 | 942 ± 126 | 21 | 2 – 45 | 2.2 ± 4.5 | = 0.0865 |

Overall, three weeks after sustaining the muscle injury, the hamstrings in the control limb showed a smaller amount of muscle volume loss when compared to the quadriceps muscle in the same leg. Therefore, the quadriceps muscle seems to be more prone to muscle loss, than the hamstrings. This is a new finding, as so far this has only been noted in immobilized patients, not in athletes (Belavý et al., 2009; Kilroe et al., 2019). Similarly, the injured hamstring muscle group lost more volume compared to the healthy leg. However, based on recent findings a volume of edema around 33.1 $cm^3$ can be expected (Crema et al., 2017). Because of resorption of edema, our observation therefore cannot safely be interpreted as an increased muscle loss in the injured hamstrings.

After three weeks of resting, the number of minutes trained correlated statistically significant inversely with the volume lost. This was the more evident case for the quadriceps muscles in both legs, while the volume loss in the injured hamstring was not statistically significant to the number of minutes trained. This may be due to edema still being resorbed, as well as an avoidance to use the injured muscle with full body load or an unaware relieving posture.

Of interest, recent literature found an increased hamstring-to-quadriceps ratio with an increasing amount of skill level and professional training in football players (Fousekis, Tsepis, & Vagenas, 2010). Additionally, low hamstring-to-quadriceps strength ratios of 0.61 or less have been found to substantially increase the risk of a muscle injury and may increase the risk of concomitant injuries such as anterior cruciate ligament (ACL) (re-) injuries (Bourne, Opar, Williams, & Shield, 2015; Croisier et al., 2008; Fousekis et al., 2010; Kellis, Galanis, & Kofotolis, 2019; Liu et al., 2012; Orchard et al., 1997; Steffen et al., 2016; Waldén, Hägglund, Werner, & Ekstrand, 2011). However, these findings have only been made in studies addressing strength ratios, not volume ratios and remain controversial (Evangelidis, Pain, & Folland, 2015; van Dyk et al., 2017). The data from this study showed that only two football players presented muscle volume ratios greater than 0.61. On average, they showed a substantially lower ratio with 0.52 ± 0.06 in the healthy leg and 0.53 ± 0.06 in the injured leg, with no relevant changes during the 6 weeks rehabilitation period. A hamstring injury therefore does not seem to have a relevant negative or a positive effect on hamstring to quadriceps volume ratios, which was not expected by the authors of this study.

It has been demonstrated that muscle volume directly correlates with strength values and muscle balance (Baxter & Piazza, 2014; Evangelidis et al., 2016). However, it is unclear if muscle volume loss of under 7% as demonstrated in our study plays a relevant role for the observed strength deficits and promote maladaptation. Change in strength, coordination and neuromuscular activity in the upper thigh in football players following a hamstring injury are common and have a proven effect on the reinjury rate (Avrillon, Hug, & Guilhem, 2020; Fyfe, Opar, Williams, & Shield, 2013; Sole, Milosavljevic, Nicholson, & Sullivan, 2011). Further studies are warranted to determine if low hamstring to quadriceps volume ratios are associated with an

**Table 2.** MRI-Analysis after 6 weeks.

| *Muscle group* | *Mean volume (MRI0) in $cm^3$ + SD* | *Mean volume ($MRI_6$) in $cm^3$ + SD* | *Mean volume loss in $cm^3$* | *95% CI* | *Mean pc in $MRI_6$ + SD* | *p-value* |
|---|---|---|---|---|---|---|
| Quadriceps injured leg | 1910 ± 191 | 1832 ± 201 | 78 | 51 – 104 | 3.8 ± 2.7 | < 0.0001 |
| Quadriceps healthy leg | 1890 ± 162 | 1807 ± 186 | 83 | 45 – 121 | 4.5 ± 3.7 | < 0.0001 |
| Hamstrings injured leg | 1004 ± 144 | 940 ± 159 | 64 | 31 – 98 | 6.5 ± 6.2 | = 0.0004 |
| Hamstrings healthy leg | 964 ± 124 | 943 ± 129 | 21 | 3 – 44 | 2.1 ± 4.5 | = 0.0956 |

increased (re-)injury risk and therefore pose clinical relevance, especially in the short-term.

This study has several limitations. The volume changes in the injured hamstrings cannot be seen as absolute changes in muscle volume, mostly due to the oedema in the injured hamstring group after trauma. Future studies need to be able to identify and control for the degree of edema consistent with a hamstring injury.

Despite evaluating their volume ratios, quadriceps and hamstring strength of the included players was not assessed. Additionally, this study provides limited standardization due to pain being the main determinant of rehabilitation progress design. Also, we did not control for type of activity, frequency, volume, intensity and programming during rehabilitation. However, this does reflect the current clinical practice, as physiotherapists treat non-professional athletes with evidence based on their individual pathology, presentation, and goals individually.

## Disclosure statement

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